Superseded Baseline Packet: pi_H = 0

Informational Power Through Pivotality

2026-05-10

Contents

1 Purpose

Superseded status note (2026-05-10). This packet should not be used as a verified result packet. A later protocol audit found that the R1 no-information-delay branch was not yet derived from a stated primitive of the extensive-form game. The majority pass branch and unanimity R2 derivation remain useful, but R1 unanimity, entry/nesting, conditional dominance, and institutional classification must be treated as pending rederivation.

This packet originally summarized the baseline architecture for the redesigned model. It is now retained as an audit artifact, not as a verified result packet.

The active manuscript `formal_model_v5.Rmd` is not the source of truth for these results. The source of truth is `model_redesign/power_architecture_derivations.Rmd`.

The baseline stacks the agenda against the hegemon:

`pi_H = 0 in Round 1 and Round 2.`

Weak states therefore control the agenda in both rounds. Any payoff advantage for the hegemon comes from veto/pivotality under unanimity and from its type-dependent outside option, not from proposal power.

2 Primitives

There are $N \geq 3$ states. One state is the hegemon H; the remaining $m = N - 1$ are weak states. Under majority, the required coalition size is

$$q = \lfloor N/2 \rfloor + 1.$$

The state is $\theta \in \{0, 1\}$. The institutional surplus is

$$V(0) = 1, \qquad V(1) = r > 1,$$

and the prior probability of the high state is μ. The expected surplus is

$$V_e(\mu) = 1 + \mu(r - 1).$$

H's outside option is external to the institutional pie and equals

$$d_H(\theta) = \alpha V(\theta).$$

Weak states have normalized outside option zero. The common discount factor is $\beta \in (0, 1)$.

Voting is sequential and public. Proposals must be feasible in the realized state in which they pass. The maintained tie-breaking rule is acceptance in indifference.

3 Result 1: Majority Pass Branch

Under majority and weak-state agenda, weak proposers can exclude H. On the pass-feasible branch

$$\mathcal{P}_M^F = \left\{ \mu : \frac{\beta(q-1)V_e(\mu)}{m} \leq 1 \right\},$$

the verified payoffs are

$$V_H^M(\mu) = \alpha V_e(\mu), \qquad V_W^M(\mu) = \frac{V_e(\mu)}{m}.$$

The majority formation set on this verified branch is

$$F_M^{pass} = \mathcal{P}_M^F \cap \left\{ \mu : \frac{V_e(\mu)}{m} \geq c \right\}.$$

This is a pass-branch result. The majority delay/rejection branch outside $\mathcal{P}_M^F$ has not yet been derived.

4 Result 2: Unanimity Round 2

Under unanimity in Round 2, a weak proposer must buy H's approval. The weak representative continuation value is

$$W_2^U(\mu) = \frac{1}{m} \max\{(1-\mu)(1-\alpha), V_e(\mu) - \alpha r\}.$$

The R2 cutoff is

$$\mu_2^* = \frac{\alpha(r-1)}{r-\alpha}.$$

Below the cutoff, the low type binds. Above the cutoff, the high type binds.

5 Result 3: Unanimity Round 1

Status: pending R1 protocol rederivation.

With $\pi_H = 0$ also in Round 2, the strict low-accepted/high-rejected R1 branch collapses into a tie under the maintained tie-breaking rule. At the candidate strict aggressive offer, the high type is indifferent rather than strictly rejecting; acceptance in indifference turns the branch into pooling.

Define

$$g(\mu) = \max\{(1-\mu)(1-\alpha), V_e(\mu) - \alpha r\}.$$

Accepted pooling in R1 pays H

$$h_P = \beta \alpha r$$

and each weak voter

$$y_P = \frac{\beta g(\mu)}{m}.$$

Pooling is feasible when

$$\beta\alpha r + \frac{(m-1)\beta g(\mu)}{m} \leq 1. \tag{U1-F}$$

The weak proposer payoff from pooling is

$$P(\mu) = V_e(\mu) - \beta\alpha r - \frac{(m-1)\beta g(\mu)}{m}.$$

Conditional on an admissible no-information delay history, the continuation payoff would be

$$R(\mu) = \frac{\beta g(\mu)}{m}.$$

This is not yet a verified option of the proposer under the original BF protocol. The following weak-proposer problem is a conditional calculation, not a theorem:

$$W^U_{1,prop}(\mu) = \max\{P(\mu) \text{ if (U1-F) holds}, R(\mu)\}.$$

Pooling is chosen when it is feasible and

$$V_e(\mu) - \beta\alpha r \geq \beta g(\mu).$$

6 Result 4: Hegemonic Rent

Status: conditional calculation, pending R1 protocol rederivation.

When accepted pooling occurs under unanimity, H receives

$$V_H^{U,P}(\mu) = \beta\alpha r.$$

Relative to H's expected outside option $\alpha V_e(\mu)$, the ex ante payoff premium is

$$\Delta_H^P(\mu) = \alpha\{\beta r - 1 - \mu(r-1)\}.$$

For $\alpha > 0$ and $\beta r > 1$, this premium is positive exactly when

$$\mu < \frac{\beta r - 1}{r - 1}.$$

This is an ex ante premium relative to the expected outside option. It does not mean that the high type receives more than its undiscounted outside option when $\beta \leq 1$.

7 Result 5: Alpha = 0 Stress Test

When $\alpha = 0$, H's outside option is no stronger than the weak states' normalized outside option. In this stress test,

$$d_H(0) = d_H(1) = 0.$$

The verified rent is zero:

$$\Delta_H(\theta, \mu) = 0.$$

Thus pivotality and private information alone do not generate positive rent in this baseline. The positive rent requires the type-dependent outside option.

8 Result 6: Entry And Nesting

Status: pending R1 protocol rederivation.

Let

$$F_U = \{\mu : V_W^U(\mu) \geq c\}.$$

On the verified majority-pass branch,

$$V_W^U(\mu) \leq V_W^M(\mu).$$

Therefore

$$F_U \cap \mathcal{P}_M^F \subseteq F_M^{pass}.$$

For the OPEC calibration, majority pass feasibility holds globally, so

$$F_U \subseteq F_M^{pass}$$

for every entry cost c. The minimum calibrated gap $V_W^M(\mu) - V_W^U(\mu)$ is 0.021375.

9 Result 7: Conditional Institutional Preference

Status: pending R1 protocol rederivation.

Conditional on both institutions forming and on the verified majority-pass branch, H's institutional comparison is

$$D_H(\mu) = V_H^U(\mu) - V_H^M(\mu).$$

For the OPEC calibration

$$N = 13, \qquad r = 1.5, \qquad \alpha = 0.19, \qquad \beta = 0.9,$$

the verified unanimity payoff is

$$V_H^U(\mu) = 0.2565 \qquad \text{for every } \mu \in [0, 1],$$

while the majority payoff on the pass branch is

$$V_H^M(\mu) = 0.19(1 + 0.5\mu).$$

Therefore

$$D_H(\mu) > 0 \iff \mu < 0.7.$$

In the OPEC calibration, conditional on both institutions forming, H strictly prefers unanimity for $\mu < 0.7$, is indifferent at $\mu = 0.7$, and strictly prefers majority for $\mu > 0.7$.

10 Result 8: Institutional Classification

Status: pending R1 protocol rederivation.

The verified general classification applies on the majority-pass domain. For the OPEC calibration, because pass feasibility holds globally, it applies on the full unit interval:

$$\begin{aligned}
&\mu \in F_U,\ \mu < 0.7 && \Rightarrow \text{H strictly chooses unanimity,}\\
&\mu \in F_U,\ \mu > 0.7 && \Rightarrow \text{H strictly chooses majority,}\\
&\mu \in F_U,\ \mu = 0.7 && \Rightarrow \text{H is indifferent,}\\
&\mu \in F_M^{pass} \setminus F_U && \Rightarrow \text{H is payoff indifferent absent a formation tie-break.}
\end{aligned}$$

This classification is narrower than the old manuscript claim. It does not say that unanimity dominates majority everywhere. It says that, in the most unfavorable agenda baseline for H, unanimity can still generate a positive payoff premium in the region where the pooling payment exceeds H's expected outside option.

11 OPEC Calibration Numbers

The verification scripts reproduce:

$$\mu_2^* = 0.072519,$$

R1 pooling feasibility through

$$\mu = 0.372424,$$

conditional dominance cutoff

$$\mu = 0.7,$$

constant unanimity payoff

$$V_H^U = 0.2565,$$

and minimum nesting gap

$$0.021375.$$

12 Verification Record

Each proof or derivation in this packet was followed by independent analytical verification. The final architecture-wide audit returned PASS without reservations.

Reproducibility scripts:

- `scripts/verify_baseline_majority_piH0.R`;
- `scripts/verify_baseline_unanimity_R2_piH0.R`;
- `scripts/verify_baseline_unanimity_R1_piH0.R`;
- `scripts/verify_baseline_stress_alpha0_piH0.R`;
- `scripts/verify_baseline_entry_nesting_piH0.R`;
- `scripts/verify_baseline_conditional_dominance_piH0.R`;
- `scripts/verify_baseline_institutional_classification_piH0.R`.

Independent verification reports:

- `quality_reports/2026-05-10_baseline_piH0_independent_verification.md`;
- `quality_reports/2026-05-10_unanimity_R1_piH0_pooling_delay_verification.md`;
- `quality_reports/2026-05-10_alpha0_piH0_stress_verification.md`;
- `quality_reports/2026-05-10_entry_nesting_piH0_verification.md`;
- `quality_reports/2026-05-10_conditional_dominance_piH0_verification.md`;
- `quality_reports/2026-05-10_institutional_classification_piH0_verification.md`;
- `quality_reports/2026-05-10_final_architecture_audit_piH0.md`.

13 Caveats To Preserve

1. Majority remains a pass-branch result. Outside $\mathcal{P}_M^F$, the majority delay/rejection branch has not been derived. The global OPEC conclusion is valid because pass feasibility holds for all $\mu \in [0, 1]$.
2. The unanimity R1 characterization uses acceptance in indifference. Without it, the aggressive low-accepted/high-rejected branch can reappear as a knife-edge selection, but not as a robust payoff-distinct branch.

3. The OPEC classification must preserve the distinction between F_U and $F_M^{pass} \setminus F_U$. In the latter region, H is payoff indifferent absent an institutional formation tie-break.

4. The positive rent is ex ante and relative to H's expected outside option.

5. The alpha = 0 stress test shows that the mechanism requires H's outside option to be type dependent.

14 Questions For External Review

1. Is the no-information delay construction a PBE under the intended sequential/public voting protocol? In particular, what beliefs are required if a proposal fails by weak-voter rejection and H's action is or is not observed?

2. Is the exclusion of the low-accepted/high-rejected R1 branch robust to all PBE-consistent off-path beliefs under acceptance in indifference?

3. Does the nesting result use the correct weak-state payoff concept under collective all-or-nothing entry, or would individual participation constraints require a different formation set?

4. Should the majority delay branch outside $\mathcal{P}_M^F$ be derived, or should the paper preserve the pass-branch domain restriction?

5. In the paper prose, should $\beta\alpha r - \alpha V_e(\mu)$ be called an informational rent, or should it be described more narrowly as an ex ante payoff premium from pooling relative to the expected outside option?